

KidLab Academy – AI & Robotics Summer Workshop

Project Overview

KidLab Academy is a full-stack workshop registration platform designed for children aged 8–14 years. The platform provides an engaging and kid-friendly environment where students and parents can learn about an AI & Robotics Summer Workshop and register online.

The application features a modern responsive interface, real-time registration handling, cloud database integration, and duplicate registration prevention.

Problem Statement

Many educational workshops rely on manual registration processes that are inefficient and difficult to manage.

The objective of this project was to create a digital platform that:

- Presents workshop information clearly.
 - Provides a child-friendly user experience.
 - Allows online registration.
 - Stores registrations securely.
 - Prevents duplicate registrations.
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Features

Kid-Friendly User Interface

- Pastel color palette
- Interactive design
- Responsive layout

- Mobile-friendly experience

Workshop Information

- Workshop overview
- Learning outcomes
- Weekly learning journey
- Key highlights
- FAQ section

Registration System

- Online registration form
- Client-side validation
- Server-side validation
- MongoDB storage

Duplicate Registration Prevention

The system checks whether the submitted email address or phone number already exists in the database before allowing registration.

Certificate Showcase

Displays a professional workshop completion certificate to increase participant confidence and engagement.

Technology Stack

Frontend

- React
- TypeScript
- Tailwind CSS
- React Hook Form
- Axios

Backend

- Node.js

- Express.js

Database

- MongoDB Atlas
- Mongoose

Deployment

- Render
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System Architecture

User → React Frontend → Express API → MongoDB Atlas

The frontend sends registration data to the backend API. The backend validates the request, checks for duplicates, stores valid records in MongoDB Atlas, and returns a response to the frontend.

Challenges Faced

- Connecting MongoDB Atlas with the deployed backend.
 - Configuring network access permissions in MongoDB Atlas.
 - Handling duplicate registrations.
 - Managing environment variables for deployment.
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Results

The project was successfully deployed and tested.

Live Application

Frontend:

<https://ai-robotics-workshop-1.onrender.com>

Backend:

<https://ai-robotics-workshop-p0g0.onrender.com>

Achievements

- Responsive user interface
 - Cloud database integration
 - Successful deployment
 - Duplicate registration prevention
 - End-to-end functionality
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Future Enhancements

- Admin Dashboard
 - Email Notifications
 - Payment Gateway Integration
 - Student Portal
 - Workshop Analytics
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Conclusion

The KidLab Academy platform successfully demonstrates full-stack web development concepts including frontend development, backend API design, database integration, form validation, cloud deployment, and user experience design. The project provides a practical solution for workshop registration while maintaining an engaging experience for children and parents.